

The Heterogeneous Effects of Government Spending

It's All About Taxes

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The Macroeconomic Puzzle

- **Empirical Discord:** Shocks yield multipliers from < 0.5 to > 1.5 .
- **Missing Variable:** Traditional models focus heavily on when spending is financed (debt vs. taxes).
- **The Overlooked Dimension:** Models largely ignore who carries the intra-temporal tax burden.

Core Hypothesis

Government spending multipliers are significantly larger when financed with **higher tax progressivity**

Historical Financing Paths (US 1913–Present)

U.S. defense spending shocks have historically relied on two distinct regimes:

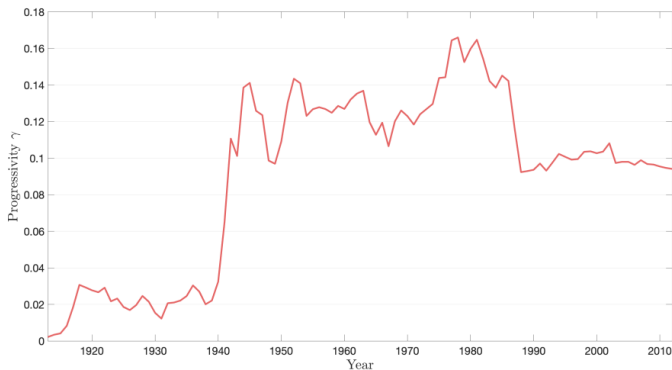
1. Progressive Financing

- WWI, WWII, Korean War.
- Burden focused on top income brackets.

2. Non-Progressive Financing

- Vietnam War, Reagan Buildup.
- Burden distributed evenly across income groups.

Tax progressivity (US 1913–Present)



The HANK Model: Structural Layers

Households

- Idiosyncratic productivity shocks (z_t).
- Wealth inequality (β -shocks).
- Indivisible labor supply choice.
- Heterogenous discount factor

Frictions: NK Blocks

- Labor Unions & Packers.
- Sticky Wages.
- Sticky Prices.

Markets: Capital & Finance

- Capital Goods Producers (ϕ^k).
- Multi-asset Financial Intermediaries.
- Clear bond and capital markets.

Policy: Government

- **Monetary:** Taylor Rule.
- **Fiscal:** Non-linear tax function:

$$\tau_l(y_l) = 1 - \lambda y_l^{-\gamma}$$

The Engines of Heterogeneity: MPC vs. LPE

Why does income level change a household's behavioral responsiveness?

Labor Participation Elasticity

- **Low-Income:** High LPE (elastic workforce entry/exit).
- **High-Income:** Low LPE (massively inelastic).

Marginal Propensity to Consume

- **Low-Income:** High MPC (liquidity constrained).
- **High-Income:** Low MPC (wealth buffers absorb shocks).

Decomposing the Consumption Response

Structural decomposition of aggregate private consumption shifts (dC):

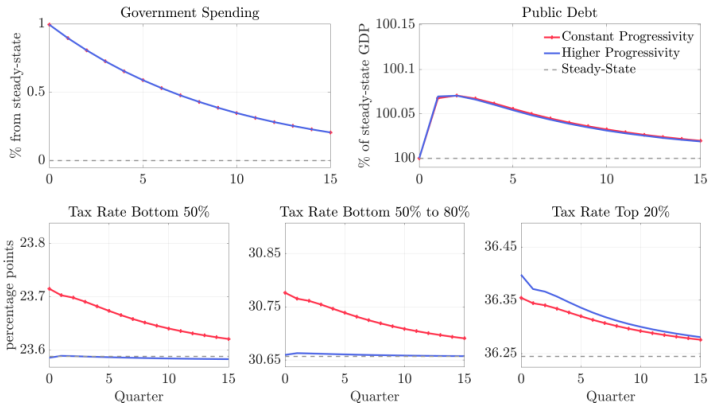
$$dC = \underbrace{-\mathbb{E}[mpc \times dT]}_{1. \text{ Tax Burden Channel}} \underbrace{-\mathbb{E}^l[mpc \times lpe^\tau \times \Delta\tau]w^hL}_{2. \text{ Labor Supply Channel}}$$

- **Tax Burden Channel:** Taxing the rich targets low- mpc agents, mitigating private consumption crowd-out.
- **Labor Supply Channel:** Taxing the rich minimizes distortions for high- lpe workers, shielding the aggregate labor supply from collapse.

Quantitative Simulation Design

- **The Shock:** +1% unexpected government spending increase ($\rho_G = 0.90$).
- **Initial Financing:** 50% debt financed ($\theta = 0.5$).
- **The Policy Counterfactual Experiments:**
 - 1 **Constant Progressivity** ($\phi = 0$): Tax rates increase identically across all income groups.
 - 2 **Higher Progressivity** ($\phi > 0$): Tax hikes are concentrated entirely on the top 20% of earners.

Fiscal Policy Transition Paths



Quantitative Findings: Multiplier Wedges

Table: Model-Simulated Cumulative Multipliers

Financing Regime	Year 1	Year 2	Year 10 (Long Run)
Constant Progressivity	0.14	0.12	0.06
Higher Progressivity	0.35	0.34	0.36

Quantitative Findings: Model IRFs

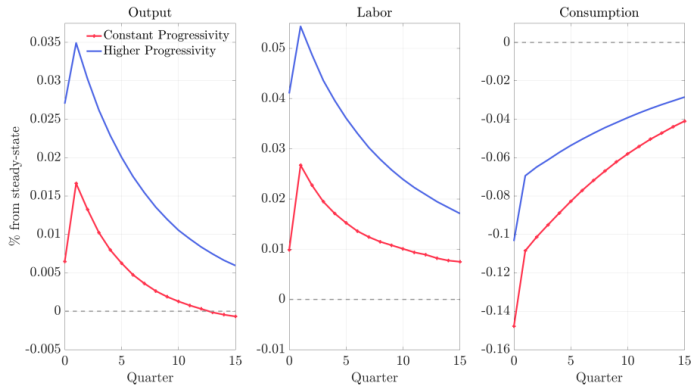


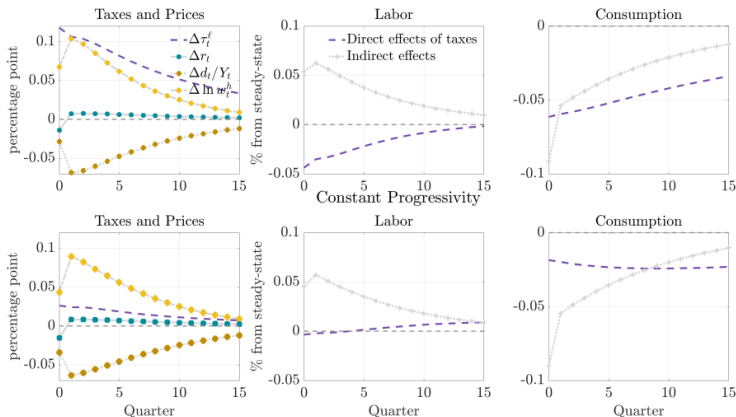
Figure 2: Model Responses to a Spending Shock Financed with Different Paths for Progressivity

Dissecting General Equilibrium Forces

What causes the structural divergence in multipliers?

- **Indirect GE Channels:** Endogenous price, wage, and dividend paths respond identically under both tax regimes.
- **Direct Allocation Channel:** Isolating household tax incidence changes replicates almost the entire multiplier wedge.
- **Takeaway:** Direct microeconomic exposure to tax shifts drives the macro multiplier gap.

Direct vs. Indirect GE Decompositions



Empirical Calibration: US Tax Responses

- **Method:** Local Projections (Jordà, 2005) on historical data (1913–2006).
- **Identification:** Military news indicators (Ramey-Zubairy).
- **Core Empirical Finding:** Historically, expansionary U.S. spending shocks raised tax rates for the top 50%, while remaining flat for the bottom 50%.
- **Implication:** The historical average baseline shock mirrors the model's progressive path.

Empirical Tax Group Responses

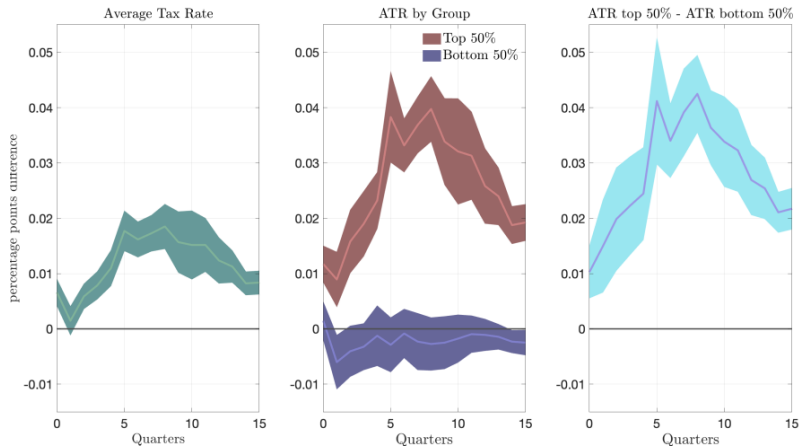


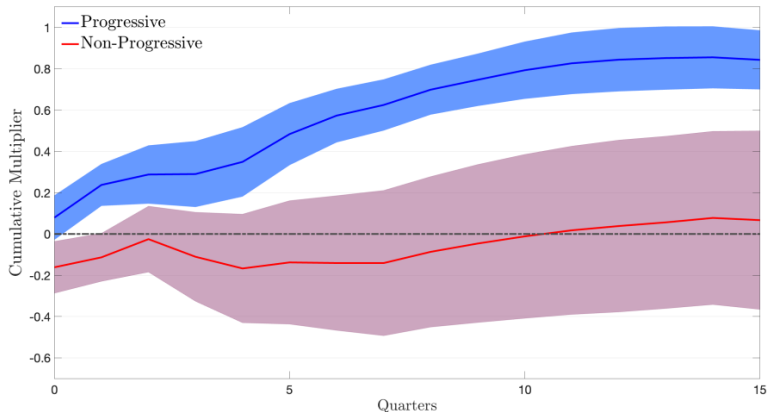
Figure 10: Average Tax Responses to a Spending Shock

Empirical Results: Conditional Multipliers

Splitting historical spending shocks via a forward-looking progressivity proxy:

- **Progressive Shocks:** Highly expansionary; cumulative output multipliers reach ~ 0.8 at Year 3.
- **Non-Progressive Shocks:** Statistically indistinguishable from zero across all time horizons.

Progressivity-Dependent Multipliers



Core Policy Takeaways

- **Primary Conclusion:** The size of the government spending multiplier is fundamentally linked to the tax incidence distribution.
- **Transmission Mechanism:** Shifting tax collection toward high earners shields the aggregate economy from consumption and labor distortion shocks.
- **Policy Prescription:** Macroeconomic evaluations of spending programs must optimize *who* bears the tax adjustment burden.